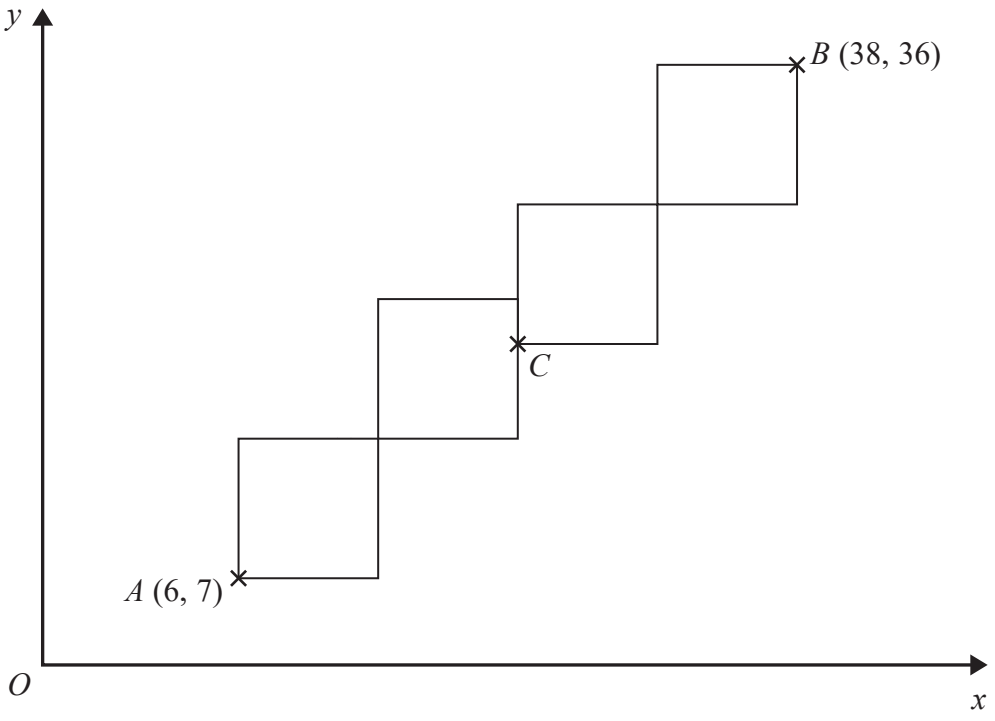
**6** A pattern is made from four identical squares.

The sides of the squares are parallel to the axes.




Point $A$ has coordinates (6, 7)
Point $B$ has coordinates (38, 36)
Point $C$ is marked on the diagram.

Work out the coordinates of $C$.

Total horizontal width = 38 − 6 = 32

**1 mark**
Find the total width

Side of each square = 32 ÷ 4 = 8

**1 mark**
Four equal horizontal widths

C is two square-widths to the right of A.

x = 6 + 2 × 8 = 22

**1 mark**
Find the x coordinate

C is two square-heights below B.

y = 36 − 2 × 8 = 20

**1 mark**
Find the y coordinate

**1 mark**
Coordinates in the order (x, y)

( 22 , 20 )

**(Total for Question 6 is 5 marks)**